Fourier-space density decay implies decay of physical-space Sobolev norms

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Editorial note. The counted statement is the author's unlabelled corollary environment at source0.tex lines 376-391; its proof environment opens at line 392, immediately after \end{corollary}, so the pairing is by adjacency and the retained proof range 392-399 is the whole environment.

Editorial note. The main theorem, with the labelled density-decay estimate T1 inside its statement (source lines 355-373), is retained as context because the counted proof applies that estimate at source line 398; the context is the complete theorem environment, which is also what supplies the label T1 referred to in the proof.

Editorial note. Because the corollary carries no author label, its identity is recorded by the pinned source position of the two environments (statement lines 376-391, proof lines 392-399) and by the claim key, which encodes that position.

Editorial note. This article carries no label-number dump and the retained counter values are left empty, so the printed numbering of the card is the wrapper's default; the original printed number of this unlabelled corollary is not recoverable from the pinned material and is therefore not asserted.

Editorial note. The retained preamble is the publisher layout adapter, the native font and layout package and the native mathematical macro and theorem block of the pinned source.

Editorial note. r002 note: the card is revived. r001 failed for two reasons that both sit on the card's side, not the author's: the material-preparation prose of this card, which the wrapper typesets, carried a literal Unicode sigma and nabla that the wrapper font does not provide, and the retained main-theorem context refers to the labelled Penrose condition display, which was not retained. r002 replaces those two characters in the card's own prose by equivalent TeX commands and retains that display as a further context. No retained author line is altered.

Editorial note. r003 note: the retained material refers to the label NH at source line 227, which was not retained; r003 adds exactly that numbered environment as a further context and changes nothing else.

Editorial scope. One original result of the pinned publisher article, counted once: physical-space decay of weighted Sobolev norms of the density (unlabelled corollary at source0.tex lines 376-391).

Allowed standard prerequisites (editorial).

- Fourier-space density decay. The Fourier transform of the density obeys a bound with a weight that grows in the frequency and in the product of frequency and time, with the exponent σ_1 fixed by the hypotheses.. It is the estimate that the counted proof transfers to physical space.
- Hausdorff-Young inequality. The Lp norm of a function is controlled by the Lp' norm of its Fourier transform, with 1/p + 1/p' = 1.. It is the first step of the proof.
- Weighted derivative norms. The norm with the weight |∇x, t∇x|^n measures derivatives in space and in the time direction simultaneously.. It is the norm appearing in the conclusion and inside the integral.

Theorem 0.1 (Nonlinear phase mixing)**.** *Fix dimension $d \geq 3$ and $M := \lceil (d+1)/2 \rceil$. Choose constants $N_0 > \sigma_1 > \sigma_0$ satisfying*

$$\sigma_1 \geq d+8, \quad N_0 - \sigma_1 > 3d/2 + 7/2 \quad \text{and} \quad \sigma_1 - \sigma_0 > d+1.$$

Let $g \in H^{N_0+7/2+\delta_0}_{2+\lceil d/2 \rceil}(\mathbb{R}^d; \mathbb{R}_+)$ for some constant $\delta_0 \in (0,1]$, and let $w \in L^1(\mathbb{R}^d, \mathbb{R})$ satisfy

$$|\widehat{w}(k)| \lesssim \frac{1}{\langle k \rangle^{M-1/2}} \tag{1}$$

for all $k \in \mathbb{R}^d$. Assume g and w satisfy the uniform Penrose condition (4) with constant c_{P}. Then there exist constants $C > 0$ and $\varepsilon_0 > 0$, depending only on $N_0, \sigma_1, \sigma_0, w, g, c_{\mathrm{P}}, d, \delta_0$, and independent of $\hbar \in (0,1]$, such that the following holds.

For any $\varepsilon \in (0, \varepsilon_0]$ and any self-adjoint initial data $Q_{\mathrm{in}} \in \mathcal{H}^{N_0}_{M,M}$ with

$$\|Q_{\mathrm{in}}\|_{\mathcal{H}^{N_0}_{M,M}} \leq \varepsilon,$$

the nonlinear Hartree equation (5) admits a unique global self-adjoint solution $Q(t)$, which satisfies:

- *Density decay: for all $k \in \mathbb{R}^d$ and $t \geq 0$,*

$$|\widehat{\rho_Q}(t,k)| \leq \frac{C\varepsilon}{\langle k, kt \rangle^{\sigma_1}}. \tag{2}$$

- *Scattering: there exists an operator $Q_\infty \in \mathcal{H}^{\sigma_0}_M(\mathbb{R}^d; \mathbb{C})$ such that, for all $t \geq 0$,*

$$\left\| e^{-i(\hbar/2)t\Delta} Q(t) e^{i(\hbar/2)t\Delta} - Q_\infty \right\|_{\mathcal{H}^{\sigma_0}_M} \leq \frac{C\varepsilon}{\langle t \rangle^{d/2}}. \tag{3}$$

$$\inf_{\substack{\mathrm{Re}\,\lambda \geq 0 \\ k \in \mathbb{R}^d}} \left| 1 + \widetilde{\mathscr{L}}(\lambda, k) \right| \geq c_{\mathrm{P}} \tag{4}$$

$$\begin{cases} i\hbar \partial_t Q = \left[-\dfrac{\hbar^2}{2}\Delta + w * \rho_Q, Q + \gamma_g \right], \\ Q(0) = Q_{\mathrm{in}}. \end{cases} \tag{5}$$

Corollary 0.2 (Physical-space decay)**.** *Let $d \geq 3$, $p \in [2, \infty]$ and let $N_0 > \sigma_1 > 0$ with*

$$\sigma_1 \geq d + 8 \quad \textit{and} \quad N_0 - \sigma_1 > 3d/2 + 7/2.$$

Then for any $n < \sigma_1 - d(1 - 1/p)$, there exist constants $C > 0$ and $\varepsilon_0 > 0$, depending only on d, p, n, N_0, σ_1, and the parameters in Theorem 0.1, and independent of $\hbar \in (0, 1]$, such that the following holds.

For any $\varepsilon \in (0, \varepsilon_0]$ and any self-adjoint initial data $Q_{\mathrm{in}} \in \mathcal{H}^{N_0}_{M,M}$ with

$$\|Q_{\mathrm{in}}\|_{\mathcal{H}^{N_0}_{M,M}} \leq \varepsilon,$$

the solution $Q(t)$ to the nonlinear Hartree equation satisfies

$$\|\langle \nabla_x, t\nabla_x \rangle^n \rho_Q(t, \cdot)\|_{L^p(\mathbb{R}^d)} \leq \frac{C\varepsilon}{\langle t \rangle^{d(1-1/p)}}. \tag{6}$$

Proof. By the Hausdorff–Young inequality,

$$\|\langle \nabla_x, t\nabla_x \rangle^n \rho_Q(t, \cdot)\|_{L^p(\mathbb{R}^d)} \lesssim_p \left(\int_{\mathbb{R}^d} \langle k, kt \rangle^{p'n} |\widehat{\rho_Q}(t, k)|^{p'} \, dk \right)^{1/p'}.$$

Using the decay estimate (2) and requiring convergence of the integral yields the result. □

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